

Introduction

Ice cover in the Great Lakes highly variable and has significant influence on regional climate, ecology, and economy. Spatially resolved ice cover ($\Delta x = 1.25\text{KM}^2$) has been recorded since 1973 and is available as daily files throughout each ice season¹. Previous work has assessed large-scale (i.e. lake or basin-wide) ice cover fluctuations through time and their relationship to global weather phenomena^{2,3,4,5,6}. We follow Mason et al. (2016)² and use the Great Lakes Aquatic Habitat Framework (GLAHF)⁷ to analyze ice cover on a sub-regional basis for the below metrics and analyze how morphometry and atmospheric conditions impact ice cover in the Great Lakes.

- **Annual Max Ice Cover (AMIC):** maximum percent during a given season
- **Seasonal Average Ice Cover (JFM):** mean daily ice cover: Jan 1 - Mar 31
- **Ice Duration:** Total (non-consecutive) days[†] where ice cover exceeded 10%

Methods

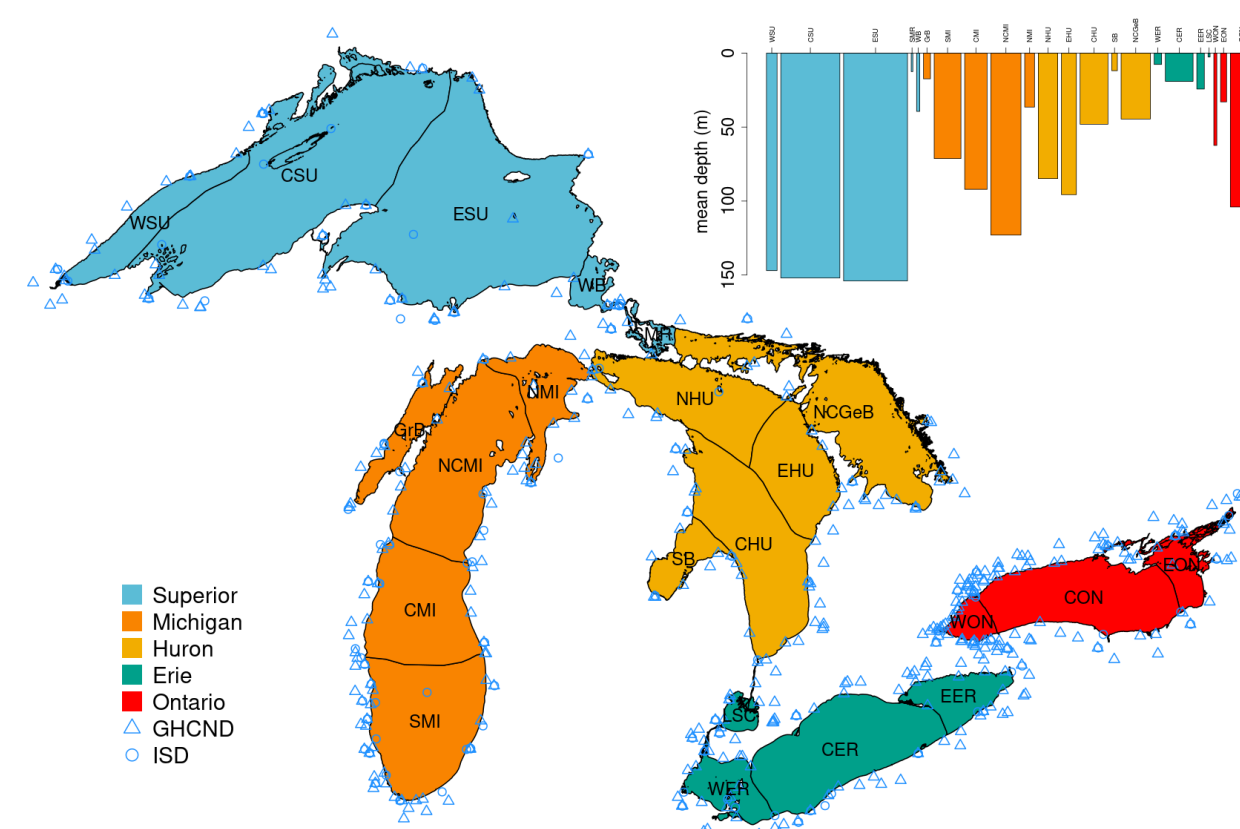


Figure 1. Spatial delineation of GLAHF subregions and selected met station locations (GHCND⁸, ISD⁹). The upper right figure shows depth and (relative) surface area of each subregion in the y and x directions, respectively.

1. Do ice trends exist at these spatial scales and how are they related to lake morphometry?
2. How do atmospheric conditions impact ice formation?

To address #1, the three metrics are calculated for each GLAHF subregion and ice season and then Mann-Kendall monotonic trend test¹⁰ is applied to check for statistical significance. Additionally correlations between trend magnitude and morphometrics are identified. To address #2, station data^{8,9} is aggregated for each subregion based on a 20 KM radius and correlations between ice metrics and monthly mean (or max for wind) meteorological surface variables are identified.

†: days are only counted Jan 1 - Mar 31 due to shorter ice reporting windows during 1970's and 80's.

Analysis

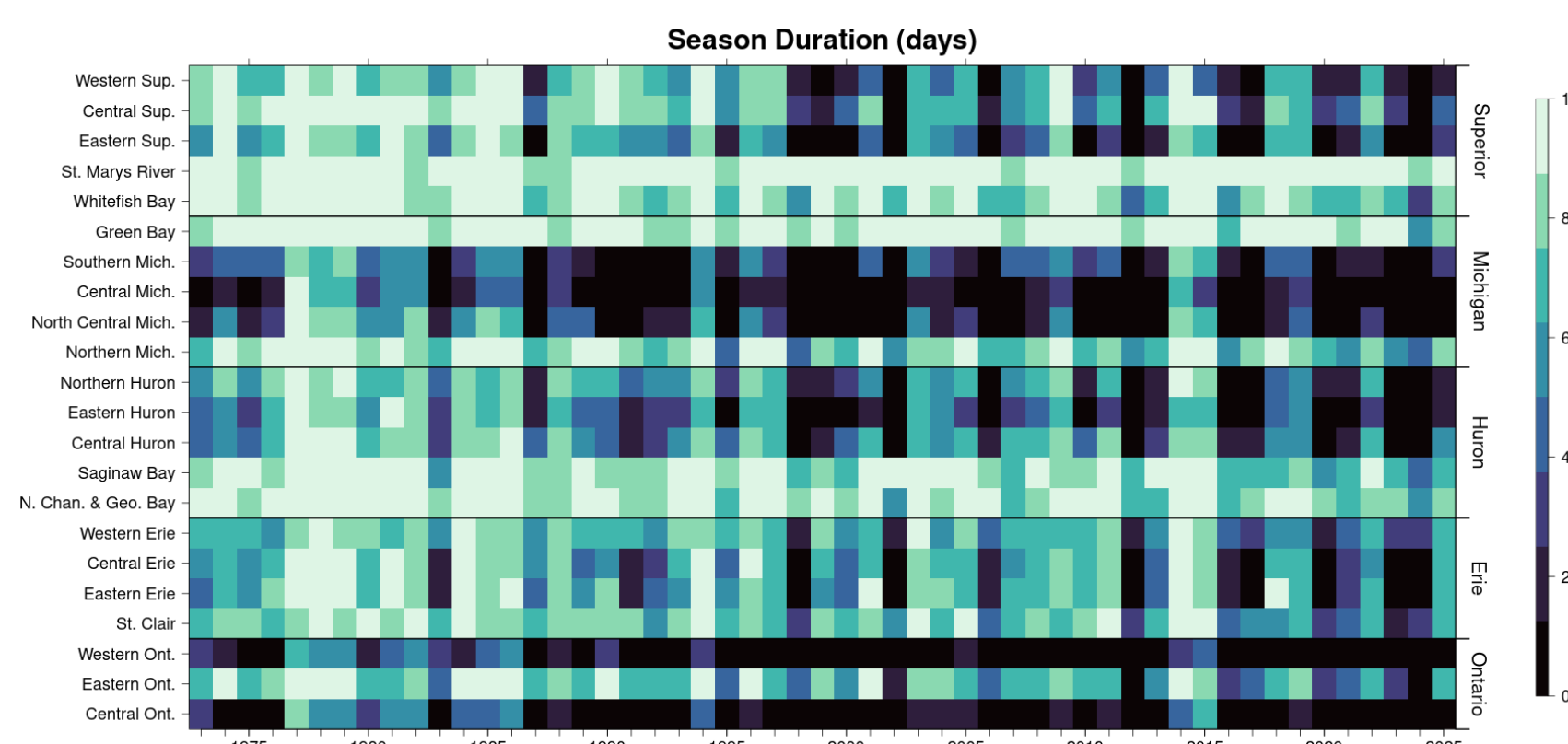


Figure 2. Evolution of one of three metrics analyzed (ice duration) for each subregion.

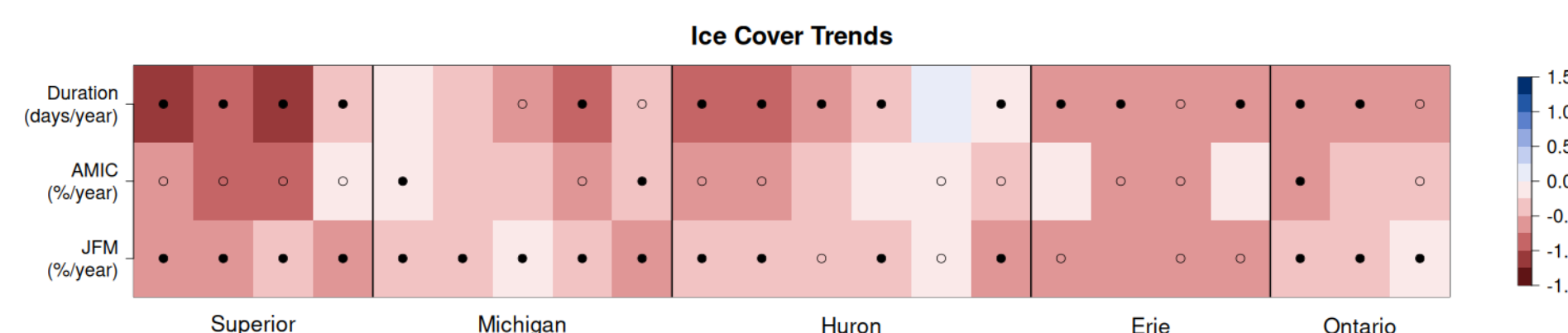


Figure 3. Linear trends for each metric. Open (solid) dots indicate 95% (99%) statistical significance. Columns correspond to rows in fig 2.

Season duration decrease is visible in some regions but not apparent in others (see fig. 2). Regions that show decrease are often offshore and deep (WSU, CSU, ESU, NHU, EHU, CHU, WON, CON). This may be due to warmer subsurface temps^{11,12} during later years providing heat to surface waters and preventing freezing, whereas shallow regions may be able to consistently release all heat from the water cc

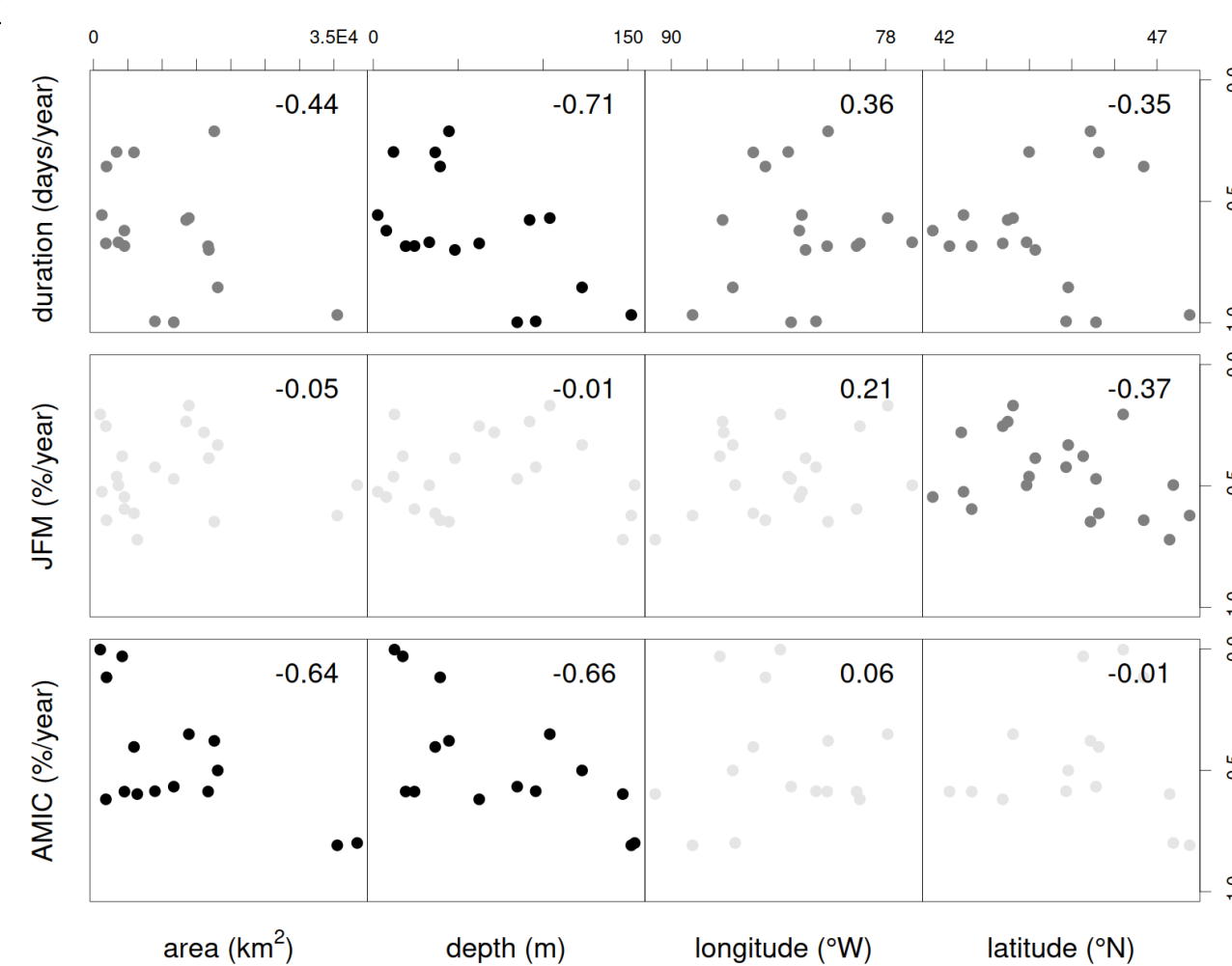


Figure 4. Correlations between statistically significant ice metric trends (vertical axes) and morphometrics (horizontal axes). Moderate correlations (Pearson's $r > 0.3$) are shown in dark grey and strong correlations (> 0.6) are black.

- depth and area show significant influence on AMIC and duration change
- latitude showed the most influence on seasonal average ice change

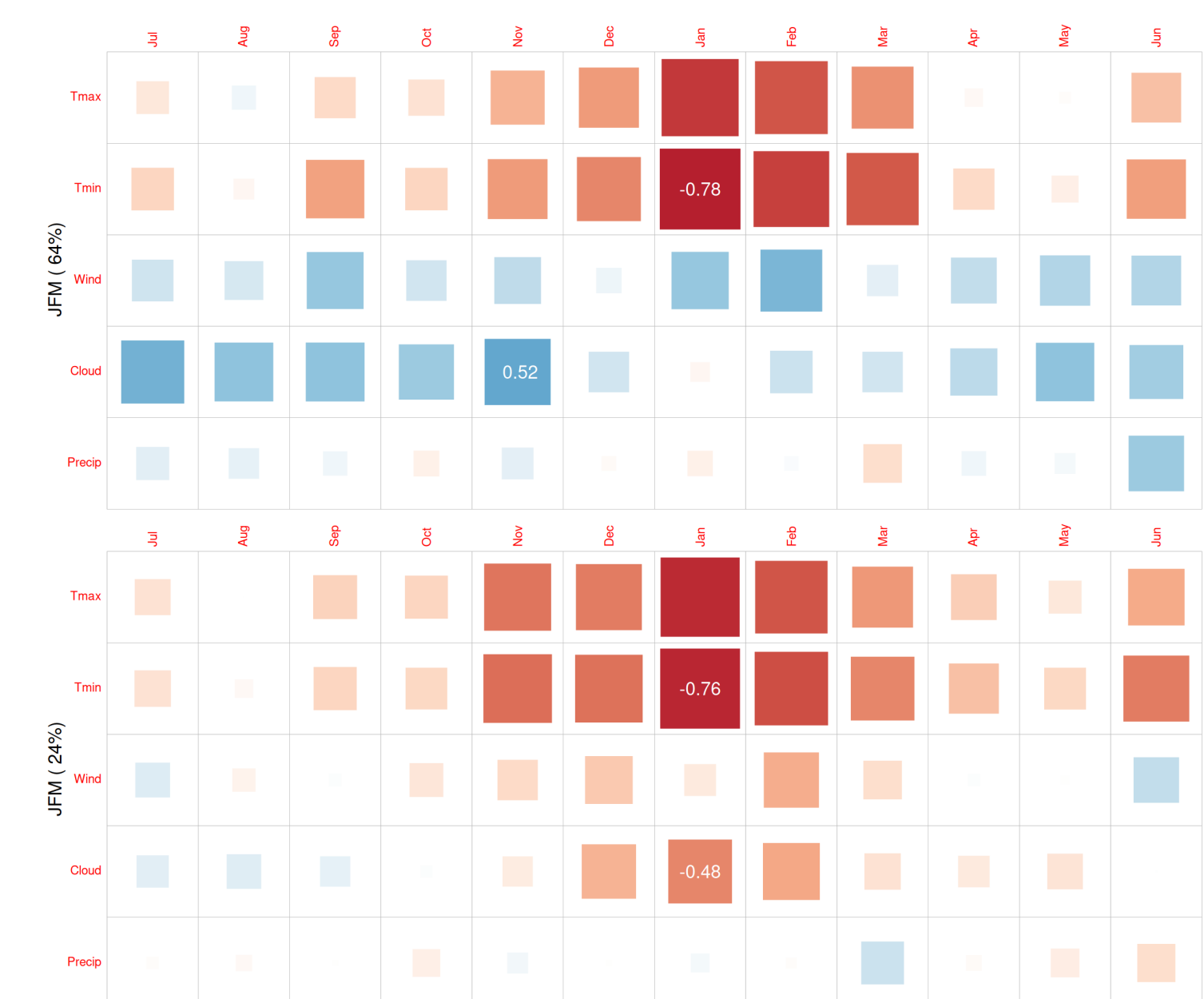


Figure 5. Correlation between Seasonal Ice Cover (JFM) and local meteorology. Meteorological variables are aggregated by month (columns) and state variable (rows). Data was correlated from Jul-Jun to each ice season during that window. Larger squares indicate stronger correlations and blue (red) indicates direct (indirect) correlation.

- Strong inverse correlation with air temp, particularly during Jan and Feb.
- Precip showed minimal correlation during any month/region while wind and cloud cover had moderate correlations.
- Since wind contributes to surface mixing which ultimately cools the water column and clouds primarily reflect solar radiation, we expect direct correlations with ice cover: True for Saginaw Bay but not Northern Huron.
- Correlations varied considerably between subregions but *within* a given subregion, the correlations were fairly consistent between AMIC, JFM and ice duration (not shown).
- A number of post ice season correlations were identified: provides valuable insight into physical processes but are not meaningful for prediction.

Summary

- A majority of subregions show significant decreasing trends in all three metrics
- Lake morphology influences ice trends; not consistent across ice metrics
- Air temperature strongly anti-correlated with ice cover for all metrics
- Wind, cloud cover correlations with ice cover varied from region to region
- Future work: predictive model development: step-wise linear regression or machine-learning



References: